

Abstract No.

HDAC Inhibition Synergizes with MEK Inhibition to Treat Senescent pLGG

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Pediatric low-grade gliomas (pLGG), the most common pediatric brain tumors, are predominantly driven by alterations activating the mitogen-activated protein kinase (MAPK) pathway, leading to oncogene-induced senescence (OIS). Although chemotherapy and MAPK-targeted inhibitors show clinical efficacy, rebound growth and disease progression remain significant challenges in unresectable tumors. Histone deacetylases (HDACs) are therapeutic targets in several cancers, including MAPK-driven tumors, but their role in pLGG remains poorly defined. Notably, we found HDAC1 expression to be increased in pilocytic astrocytoma compared with normal cerebellum. Given the clinical availability of HDAC inhibitors, we investigated their anti-tumor activity and their potential as combination partners for established therapies in pLGG.

A panel of clinically relevant HDAC inhibitors was tested in drug sensitivity and combination screens using patient-derived pLGG cell lines harboring KIAA1549::BRAF fusions or BRAF-V600E alterations in proliferation and OIS. Drug responses were assessed using metabolic activity assays, viable cell counts, and caspase-3/7 activity measurements. Synergy was quantified using the Bliss model on the platform SynergyFinder. Effects on gene expression and protein abundance were investigated by qPCR and western blot, respectively.

Several HDAC inhibitors showed anti-tumor activity across pLGG models, with romidepsin and fimepinostat, both targeting class I HDACs including HDAC1, emerging as the most promising compounds active at clinically achievable concentrations. Combination screening identified an in-class synergy between HDAC inhibitors and MEK inhibitors. Bliss synergy scores were particularly high in senescent cells, especially in KIAA1549::BRAF fusion models. The combination of romidepsin and trametinib significantly reduced viable cell numbers and induced apoptosis, indicated by increased caspase-3/7 activity. Mechanistically, co-treatment was associated with accumulation of the pro-apoptotic BH3-only proteins BIM and BCL-2 and transcriptional changes consistent with a shift towards a pro-apoptotic state.



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In conclusion, HDACi-MEKi co-treatment may represent a novel strategy to improve MAPK-targeted therapy in pLGG, particularly by enhancing the elimination of senescent tumor cells.



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